Mixed-Initiative Orchestration of a Managed-Agent Catalog: Intent Mining and Provenance-Gated Narration in a Human-AI Chat Console

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1. Motivation and Gap

Horvitz's twelve principles for mixed-initiative user interfaces couple interface agents with direct manipulation rather than opposing them, and call for considering uncertainty about a user's goals, resolving uncertainty through dialog, maintaining a working memory of recent interactions, and continuing to learn by observing [6]. That framework assumed one intelligent service collaborating with one user to infer and achieve that user's goals [6].

Current language-model agent systems have not preserved this coupling. Fully autonomous language-model agents face limited reliability from hallucination, difficulty on complex tasks, and substantial safety risk, and human feedback and control are reintroduced specifically to mitigate these failures [8]. At the interaction layer, current chatbots, copilots, and agents remain siloed, which forces users to restate intent and track dependencies across tools [12]. Both name the same unresolved element, a shared and inspectable way to carry user intent across a system of agents rather than within one [8, 12].

Contemporary systems instead expose a catalog of agents with different authorities and skills, coordinated by an orchestrator, where agency is negotiated through control between human and machine agents rather than assigned as a fixed property of either [4]. The open question is how Horvitz's principles extend to that setting, where intent must be mined continuously from a chat console, relayed across authority tiers, and where every narrated value is bound to the evidence that produced it. We study this with a deployed system in which routing is only the demonstration domain.

2. Background

Retrieval-augmented generation combines parametric and non-parametric memory, and its originating work states that providing provenance for model decisions remains an open research problem [7]. Tool use interleaves reasoning traces with tool actions to yield interpretable decision traces [13], and trains models to decide which interfaces to call, when, and with what arguments [11]. The Attributable to Identified Sources framework defines attribution as the accurate use of source documents to support generated text and links it to hallucination and faithfulness [10]. Perceived shared understanding can track interaction fluency even when actual shared understanding is limited, so narrated values must be checked against identified sources rather than inferred from conversational quality [5].

The pax console composes these components. A registry holds a catalog of managed agents, each with an authority tier of public, researcher, or keyed and a tool set. A routing orchestrator mines user intent as a structured record of slot, method, place, and resolved location, and returns a directive naming the active slot, the interaction node, and a prompt. Selecting the next interaction node is an affordance relation between user and console, defined by threshold criteria rather than a fixed feature of any single agent [2]. Under a prepare-then-narrate contract the model may state only values returned by tools, and each value carries a provenance label of live, partially live, or deferred. These labels instantiate the entity-activity-agent provenance model [9] and the distinction between retrospective provenance, an execution trace from the current request, and prospective-only provenance, a recipe whose execution is not present in the request [3]. The architecture is a candidate instantiation of the mixed-initiative principles in the multi-agent setting [6].

3. Research Goal and Hypothesis

An orchestrator that mines user intent from a chat console and relays it across a catalog of managed agents, with every narrated value gated by its provenance, is hypothesized to satisfy the mixed-initiative principles better than a single-agent or fully autonomous baseline [6]. Provenance-gated orchestrated intent relay should lower the unsupported-claim rate and reduce intent restatements relative to a single-agent baseline, without raising slot-filling latency beyond an acceptable bound. The negative control bypasses the orchestrator so the console behaves as a single agent.

4. Proposed Experimental Approach

The deployed pax console is the testbed, with routing as the task domain. Three conditions share a task set, tool layer, and retrieval index: a single-agent console with one concierge and no orchestrator, an autonomous multi-agent console without human-in-the-loop intent confirmation, and the orchestrated mixed-initiative console.

Each session records four measures. The unsupported-claim rate is scored with the Attributable to Identified Sources protocol, marking each narrated statement as supported or unsupported by the tool result it cites [10]. Intent-restatement count records how often the user re-enters a goal the system already held, the burden named in the unified-architecture analysis [12]. Slot-filling latency and error rate measure interaction cost. Provenance-label accuracy checks that each live, partially live, or deferred label matches whether a tool read occurred in the request, following the retrospective and prospective provenance distinction [3]. This reports transparency about what the system executed, distinct from the explainability of the narration [1]. Holding the retrieval configuration [7] and the tool-invocation policy [13, 11] constant isolates the effect of the orchestration and the provenance gate.

5. Expected Outcomes and Significance

If the hypothesis holds, the orchestrated console would show a lower unsupported-claim rate than both baselines, because the prepare-then-narrate contract and the provenance gate block any statement not backed by a tool read [10], and fewer intent restatements than the single-agent console, because mined intent is carried across agents rather than re-elicited [12]. These results map onto the mixed-initiative principles of resolving uncertainty through dialog and maintaining a working memory of recent interactions [6]. If orchestration instead raises latency without reducing unsupported claims or restatements, the single-agent console suffices and the added coordination is not justified.

The significance is a concrete extension of the mixed-initiative principles from a single agent to an orchestrated catalog of agents with differing authority, addressing the unsolved element named in recent human-agent systems work [8, 12]. The provenance gate makes the result auditable, since every compared value is marked as a live tool read or a deferred estimate [9, 3]. The scope is bounded to one console, one task domain, and one tool layer, with no claim about other domains or model families. Routing is the demonstration surface, not the contribution, which is the intent-relay and provenance-gate protocol evaluated against principles established for the single-agent case [6].

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